

Spoken Tutorial – Introduction to AI, ML, and DL

Title: Introduction to AI, ML, and DL

Module	Artificial Intelligence Basics
Episode	1
Learning Objective	At the end of this tutorial learner will be able to 1. Define Artificial Intelligence (AI). 2. Explain Machine Learning (ML) and its relation to AI. 3. Compare traditional programming with ML. 4. Describe Deep Learning (DL) and its applications.
Approx. Duration	15 minutes
Outline	• Introduction to Artificial Intelligence (AI) • Defining Machine Learning (ML) • Traditional Programming vs. Machine Learning Analogy • Understanding Deep Learning (DL) • Applications of Deep Learning
Meta Tags	AI, Artificial Intelligence, Machine Learning, ML, Deep Learning, DL, Neural Networks, Computer Science, Spoken Tutorial
Pre-requisite Tutorial	Basic computer literacy. No prior coding knowledge is needed.

Script

Visual Cue	Narration
Title slide for a tutorial on AI, ML, and DL, with a futuristic, clean design.	Welcome to this Spoken Tutorial on Introduction to AI, ML, and DL.
Slide showing bullet points for learning objectives, with icons representing learning and goals.	In this tutorial, you will learn, What Artificial Intelligence is. How Machine Learning relates to AI. The difference between traditional programming and ML. What Deep Learning is and how it works.
Slide showing a computer monitor and a mobile phone, indicating system requirements.	Here, I am using a browser on a computer or mobile.
Slide with icons for internet connectivity and 'no coding required'.	To follow this tutorial, you must be connected to the internet. No coding is needed for this tutorial.

Conceptual image of artificial intelligence, brain-like circuits, robots, and data flows.	Let's begin by understanding Artificial Intelligence. AI is a broad field of computer science. Its goal is to build smart machines. These machines perform tasks needing human intelligence. This includes understanding language and recognizing images. AI can also solve complex problems. The idea of AI started in the 1950s. Pioneers asked, 'Can machines think?' AI is the entire universe of making computers intelligent.
Image depicting machine learning, with data points, patterns, and a learning algorithm.	Now, let's look at Machine Learning. Machine Learning is a type of AI. Instead of rules, we give it a lot of data. The computer learns rules for itself. It is all about recognizing patterns from examples.
Illustration of a cat with explicit labels for ears, whiskers, fur, and legs, representing rule-based programming.	Let's use an analogy to understand better. Imagine programming a computer to identify a cat. In traditional programming, you give rules first. You would write: IF it has pointy ears, AND whiskers, AND fur, AND four legs, THEN it is a cat. What happens if the cat's ears are folded? This method is fragile and can fail easily.
Image of many different cat pictures being fed into a machine learning model, showing pattern recognition.	Now consider Machine Learning, which is data first. You show the computer thousands of cat pictures. Each picture is labeled as 'cat'. The ML model analyzes all these images. It learns common patterns and features by itself. It builds its own internal rules. These rules are much more flexible and accurate.
Diagram of a multi-layered neural network, inspired by the human brain, showing 'deep' structure.	Next, let's understand Deep Learning. Deep Learning is an advanced version of Machine Learning. It uses a structure called a neural network. This network is inspired by the human brain. Neural networks have many layers of 'neurons'. Because there are many layers, we call it 'deep'. This deep structure learns complex patterns from data. It is more powerful than traditional ML.
Collage of images representing deep learning applications: self-driving car, voice assistant, generative AI art.	Deep Learning is behind many modern applications. It helps self-driving cars recognize pedestrians. Voice assistants understand your commands. It is also used in generative AI. This is the magic behind many smart technologies.
Summary slide with key takeaways about AI, ML, and DL in bullet points, with a checkmark icon.	Let's summarize what we learned. AI is a broad field for making machines intelligent. ML is a type of AI that learns from data. Deep Learning is advanced ML using neural networks. It learns complex patterns with many layers.

Assignment slide with a question mark and a notebook, suggesting an activity.	Now, let's do a small activity. Think of a simple task, like 'sorting laundry'. Write down how you would program it traditionally. Then, describe how an ML model would learn to do it. Consider what data you would provide for ML.
EduPyramids Educational Services Private Limited logo, SINE, IIT Bombay logo, and 'Thank you' message.	This brings us to the end of this tutorial. This Spoken Tutorial is brought to you by EduPyramids Educational Services Private Limited, SINE, IIT Bombay. Thank you for joining!